

7. Natural gas is a source of methane.

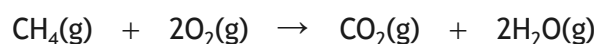
(a) Methane, CH₄, can be used as a fuel.

In an experiment, methane was burned to raise the temperature of 100 cm³ of water by 27 °C.

Using the enthalpy of combustion of methane (891 kJ mol⁻¹), calculate the mass of methane, in g, burned in this experiment.

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(b) The equation for the combustion of methane is shown.



Bond enthalpies can be used to calculate a theoretical enthalpy change for this reaction.

Using bond enthalpies from the data booklet, calculate the enthalpy change, in kJ mol⁻¹, for the combustion of methane.

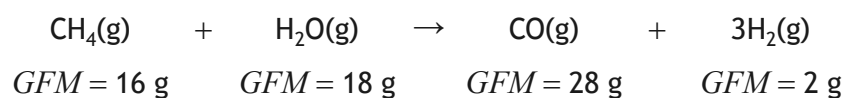
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* X 8 1 3 7 6 0 1 1 8 *

7. (continued)

(c) Methane reacts with steam to produce hydrogen.



Calculate the atom economy for the formation of hydrogen.

2

(d) Another naturally occurring gas is nitrogen dioxide, NO_2 .

Nitrogen dioxide exists in equilibrium with dinitrogen tetroxide, N_2O_4 .



Complete the table to show the conditions that would maximise the yield of nitrogen dioxide.

1

Condition	High/Low
Temperature	
Pressure	

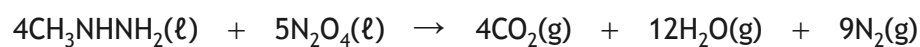
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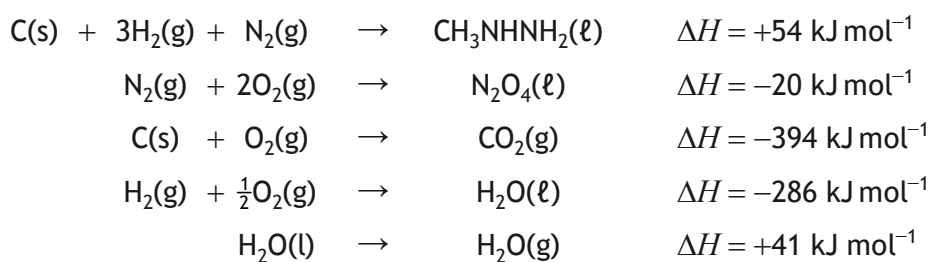
7. (continued)

- (e) (i) In the United States Space Shuttle, dinitrogen tetroxide was reacted with methylhydrazine.



Calculate the enthalpy of this reaction, in kJ, by using the data shown below.

2



- (ii) Draw the full structural formula for methylhydrazine, CH_3NHNH_2 .

1



* X 8 1 3 7 6 0 1 2 0 *